value of one widget changes, its valueChanged(int) signal -is emitted, and the setValue(int) slot of the other widget is called with the new value.

spinBox->setValue(1947) sets the spin box value to 1947. When this happens, the QSpinBox emits the valueChanged(int) signal with an int argument of 1947. This argument is passed to the QSlider's setValue(int) slot, which sets the slider value to 1947. The slider then emits the valueChanged(int) signal because its own value changed, triggering the spin box's setValue(int) slot. But at this point, setValue(int) doesn't emit any signal, since the spin box value is already 1947. This prevents infinite recursion. In short, changing one widget's value changes another.

One interesting thing to note here is that even though we didn't set the position or size of any widget explicitly, the QSpinBox and QSlider appear nicely laid out side by side. This is because QHBoxLayout automatically assigns reasonable positions and sizes to the widgets for which it is responsible, based on their needs.

Qt's approach to building user interfaces is simple to understand and very flexible. The most common pattern that Qt programmers use is to instantiate the required widgets and then set their properties as necessary. Programmers add the widgets to layouts, which automatically take care of sizing and positioning.

Simple QDialog

Now that we are done with the basics, let's create a QDialog.

Most GUI applications consist of a main window with a menu bar and toolbar, along with dozens of dialogs that complement the main window. It is also possible to create dialog applications that respond directly to the user's choices by performing the appropriate actions (e.g., a calculator application).

We will create our first dialog purely by writing code to show how it is done. Then we will see how to build dialogs using Qt Creator, Qt's visual design tool. Using Qt Creator is a lot faster than hand-coding and makes it easy to test different designs and to change designs later.

The Find Dialog

The source code is spread across two files: finddialog.h and finddialog.cpp.

finddialog.h.

```
#ifndef FINDDIALOG_H
```